ENTRY 036 Floor-Collapse Is the Independence Breakdown, Amplified — August 16, 2026 — Status: COMPLETE

Entry 036 — Every Channel Looks Fine in Isolation, Combined They Erase the Circuit

Three Fresh Instances, None Involving Qubits 5/6, Same Signature as Entries 029 and 033 -- Just Far Stronger and on Much Shorter Routes

1. Method

Three floor-collapse instances were picked from the combined Entry 034/035 outlier pool, deliberately avoiding qubits 5 and 6 (Entry 035 already showed they aren't special) and choosing cases where v4.1 predicted well above the floor -- genuine misses, not places the model already (correctly) predicted near-50%. Each was isolated against Aer into gate-error-only, decoherence-only, and readout-only noise models, the same ablation used in Entries 028, 029, and 033.

2. Result: Every Channel Checks Out, the Combination Doesn't

route	edges	gate err.	decoherence	readout	naive product	REAL combined	v4.1 predicts
Sherbrooke 3→6	3	97.7%	97.7%	91.9%	87.8%	50.0%	88.5%
Sherbrooke 40→56	5	93.3%	84.6%	90.8%	71.6%	50.7%	81.9%
Kyiv 95→72	8	90.4%	71.5%	99.4%	64.2%	50.0%	60.9%

In every instance, each individual channel isolated against Aer looks completely ordinary -- 72-98% survival, nothing alarming. Multiplying even Aer's own perfectly isolated per-channel results together lands at 64-88%, matching v4.1's prediction closely (as expected, since v4.1 is built on the same multiplicative-composition assumption). But the REAL combined-noise result on all three routes sits within 1 point of the 50% floor -- total entanglement failure, with nothing in any single channel foreshadowing it.

3. This Is Entries 029/033's Finding -- Just Much Stronger, on Much Shorter Routes

Entry 029 found an 11-point channel-independence gap on Kyiv's worst-T1 route. Entry 033 found a 22-point gap on Sherbrooke's longest route (25 SWAP legs). This entry's gaps are 37.8, 21.0, and 14.2 points -- comparable to or LARGER than Entry 033's, but on routes of just 3, 5, and 8 edges. Entry 033 concluded the breakdown was driven by "accumulated exposure," implicitly assuming more hops means more correlation. A 3-edge route producing the single largest independence gap measured anywhere in this project rules that assumption out: route length is not the variable that predicts how badly the channels interact. Something about which specific edges combine -- not how many -- drives the correlation strength.

Entry 036 — Conclusion

Floor-collapse is not a new bug and not a qubit defect -- it is the channel-independence breakdown documented in Entries 029 and 033, occurring far more severely than previously seen, and on routes short enough that route length alone cannot be the predictor. This raises the stakes on the limitation Entry 033 flagged but set aside: it isn't just costing a few points on the two longest routes per chip, it is erasing 14.5% of Sherbrooke's short routes completely. The multiplicative-composition framework this project's closed forms (Entries 025, 028, 032) are built on has a real, now well-evidenced ceiling, and no closed-form patch discovered so far reaches past it.

4. Where This Leaves the Project

Fixing this properly requires modeling correlated noise across a route rather than composing independently-measured channel probabilities -- effectively a step toward simulating the density matrix along the path instead of multiplying survival probabilities, which Entry 033 already identified as the real fix and set aside as out of scope for a closed-form model. That assessment stands. What changes here is urgency and framing: this is exactly the kind of structured, high-value target a learned model (GNN or otherwise) could plausibly pick up where the closed-form approach hits its ceiling -- the batch dataset from Entries 034/035 already contains 12 labeled floor-collapse examples for it to learn from directly.

5. Next Step

Move into the ML/GNN phase using the Entry 034/035 batch dataset (238 circuits combined) as the training/validation set, with v4.1's predictions as the closed-form benchmark any learned model needs to beat -- particularly on the floor-collapse cases this entry shows the closed-form approach structurally cannot reach.

Research Log — Index Addendum (Entry 036)

Entry	Title	Date	Status
Entry 036	Floor-Collapse Is the Independence Breakdown, Amplified	Aug 16, 2026	Complete

Script: entry036_isolate_floor_collapse.py. Data: quantumbridge_data/entry036_isolate_floor_collapse.json. Confirms floor-collapse is Entries 029/033's channel-independence breakdown, stronger and on shorter routes than previously seen -- rules out route length as the predictor. No code changes; closes the floor-collapse investigation opened in Entry 034. Next: ML/GNN phase.